

Fast ICA Algorithm

BSS, FICA and CORDIC

Nimal
Praful
Chanakya

The Problem: Blind Source Separation (BSS)

What is it?

BSS aims to recover original source signals (S) from a set of observed mixed signals (X). Think of isolating a single speaker's voice from a noisy room (the "cocktail party problem").

The Problem: Blind Source Separation (BSS)

What is it?

BSS aims to recover original source signals (S) from a set of observed mixed signals (X). Think of isolating a single speaker's voice from a noisy room (the "cocktail party problem").

The "Blind" Challenge

This is done **without** prior knowledge of the original sources or how they were mixed.

The Mathematical Model

The basic linear mixing process is represented as:

$$X = AS$$

- X : Mixed signals
- A : Unknown mixing matrix
- S : Original signals

The Method: ICA and the FastICA Algorithm

Independent Component Analysis (ICA)

ICA is the most popular statistical method for solving the BSS problem.

- **Goal:** Find an **unmixing matrix** (W), which is essentially the inverse of the mixing matrix (A^{-1}).
- **Recovery:** The estimated sources (S_{est}) are then found by:

$$S_{est} = W^T Z$$

(Where Z represents preprocessed, whitened data).

The Method: ICA and the FastICA Algorithm

Independent Component Analysis (ICA)

ICA is the most popular statistical method for solving the BSS problem.

- **Goal:** Find an **unmixing matrix** (W), which is essentially the inverse of the mixing matrix (A^{-1}).
- **Recovery:** The estimated sources (S_{est}) are then found by:

$$S_{est} = W^T Z$$

(Where Z represents preprocessed, whitened data).

FastICA (FICA): The Popular Choice

- **Pros:** Known for its **high convergence speed** and accuracy.
- **Cons:** **Computationally intensive!** The algorithm requires many costly arithmetic operations (division, square roots, matrix operations), making it unsuitable for low-power, resource-constrained applications like mobile health sensors.

The Tool: CORDIC to the Rescue!

The Challenge with FICA's Hardware Implementation

Directly implementing FICA's iterative update step requires complex, power-hungry hardware for operations like division and square root evaluation.

The Tool: CORDIC to the Rescue!

The Challenge with FICA's Hardware Implementation

Directly implementing FICA's iterative update step requires complex, power-hungry hardware for operations like division and square root evaluation.

The Solution: CORDIC (COordinate Rotation DIgital Computer)

CORDIC is a hardware-efficient algorithm perfect for vector rotations and trigonometric calculations.

- **Superpower:** It avoids complex multipliers and instead relies only on simple **shift-and-add** operations.
- **Modes:**
 - 1 **Rotation Mode:** Rotates a vector by a known angle θ .
 - 2 **Vectoring Mode:** Finds the angle and magnitude of a given vector.

The Big Idea: A Unified CORDIC Architecture

The core innovation is to **reuse a single CORDIC unit** for nearly all stages of the FICA algorithm, not just for the standard preprocessing step.

FICA Steps → Mapped to CORDIC Operations

- **FICA Iteration:** The update rule's math is geometrically equivalent to a CORDIC rotation.

The Big Idea: A Unified CORDIC Architecture

The core innovation is to **reuse a single CORDIC unit** for nearly all stages of the FICA algorithm, not just for the standard preprocessing step.

FICA Steps → Mapped to CORDIC Operations

- **FICA Iteration:** The update rule's math is geometrically equivalent to a CORDIC rotation.
- **Normalization:** A vector is normalized by finding its angle (Vectoring Mode) and then generating sine/cosine values (Rotation Mode).

The Big Idea: A Unified CORDIC Architecture

The core innovation is to **reuse a single CORDIC unit** for nearly all stages of the FICA algorithm, not just for the standard preprocessing step.

FICA Steps → Mapped to CORDIC Operations

- **FICA Iteration:** The update rule's math is geometrically equivalent to a CORDIC rotation.
- **Normalization:** A vector is normalized by finding its angle (Vectoring Mode) and then generating sine/cosine values (Rotation Mode).
- **Component Estimation:** The final step, another dot product, is also performed as a CORDIC rotation.

Conclusion: Smart Design for Smart Devices

- **Problem:** Conventional FICA is too computationally expensive for low-power, mobile applications (e.g., wireless sensors, wearable health monitors).

Conclusion: Smart Design for Smart Devices

- **Problem:** Conventional FICA is too computationally expensive for low-power, mobile applications (e.g., wireless sensors, wearable health monitors).
- **Solution:** An innovative algorithm-architecture co-design that replaces complex arithmetic hardware with a single, reusable CORDIC unit.

Conclusion: Smart Design for Smart Devices

- **Problem:** Conventional FICA is too computationally expensive for low-power, mobile applications (e.g., wireless sensors, wearable health monitors).
- **Solution:** An innovative algorithm-architecture co-design that replaces complex arithmetic hardware with a single, reusable CORDIC unit.
- **Key Benefits:**
 - Dramatically reduces hardware complexity, area, and power consumption.
 - Intelligently maps the FICA algorithm's geometry onto the CORDIC architecture.
 - Makes advanced signal processing feasible on resource-constrained hardware.